

Induction, Recursion & Recurrences

Where is the logical bridge?

COMSC 2043

Tuesday, October 6, 2026

Today's question

Where is the logical bridge—and are we assuming what we are trying to prove?

Sources → Rules/Assumptions → Work → Check → Summary

A polished function that never stops

```
def S(n):  
    return n + S(n - 1)
```

What does $S(3)$ do?

The missing stop

$$S(3) \rightarrow S(2) \rightarrow S(1) \rightarrow S(0) \rightarrow S(-1) \rightarrow \dots$$

Takeaway

A recursive definition needs a place to stop, and each step must move toward it.

Demo 0: a terminating sum

```
def S(n):  
    if n == 0:                # base case  
        return 0  
    return n + S(n - 1)      # reduce toward 0
```

Base case: $S(0) = 0$ Step: $S(n) = n + S(n - 1)$

Trace before you trust it

$$\begin{aligned} S(3) &= 3 + S(2) \\ &= 3 + (2 + S(1)) \\ &= 3 + (2 + (1 + S(0))) \\ &= 3 + (2 + (1 + 0)) = 6 \end{aligned}$$

Each call lowers a nonnegative integer, so the trace reaches 0.

A definition computes; a proof earns the formula

- ▶ The recursive rule tells us how to compute $S(n)$.
- ▶ Matching a formula for 30 inputs is evidence about those inputs.
- ▶ Induction shows why the match continues for every $n \geq 0$.

$$\text{Claim: } S(n) = \frac{n(n+1)}{2}$$

Sum proof: Base

Base case: $n = 0$

$$S(0) = 0 = \frac{0(0+1)}{2}.$$

The claim starts true.

Sum proof: Hypothesis

Inductive hypothesis

For one fixed $k \geq 0$, assume $S(k) = \frac{k(k+1)}{2}$.

We assume the k case so we can earn the next case.

Sum proof: Step

$$\begin{aligned} S(k+1) &= (k+1) + S(k) \\ &= (k+1) + \frac{k(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Takeaway: the hypothesis bridges k to $k+1$.

Demo 1: Fibonacci needs two starting values

$$F(0) = 0,$$

$$F(n) = F(n-1) + F(n-2),$$

$$F(1) = 1,$$

$$n \geq 2.$$

Why two?

$F(2)$ reaches back to both $F(1)$ and $F(0)$.

The recursive definition

```
def fib(n):  
    if n == 0:  
        return 0                # base case 1  
    if n == 1:  
        return 1                # base case 2  
    return fib(n - 1) + fib(n - 2)
```

Trace $F(5)$

$$F(2) = F(1) + F(0) = 1 + 0 = 1$$

$$F(3) = F(2) + F(1) = 1 + 1 = 2$$

$$F(4) = F(3) + F(2) = 2 + 1 = 3$$

$$F(5) = F(4) + F(3) = 3 + 2 = 5$$

0, 1, 1, 2, 3, 5

Termination is not the same as correctness

- ▶ Missing $F(1)$: one branch runs through negative inputs; the guard raises `RecursionError`.
- ▶ Wrong $F(1) = 0$: the calls stop, but the first six values are $[0, 0, 0, 0, 0, 0]$.
- ▶ The step recombines values; the base cases supply them.

Check both base cases—and their values.

Fibonacci proof: Base

Claim

$F(1) + \cdots + F(n) = F(n+2) - 1$ for $n \geq 0$; the empty sum is 0.

Base case: $n = 0$

$$0 = F(2) - 1 = 1 - 1.$$

Fibonacci proof: Hypothesis

Assume for one $k \geq 0$ that

$$F(1) + \cdots + F(k) = F(k+2) - 1.$$

The goal is the sum through $F(k+1)$.

Fibonacci proof: Step

$$\begin{aligned} F(1) + \cdots + F(k+1) &= (F(k+2) - 1) + F(k+1) \\ &= F(k+3) - 1. \end{aligned}$$

Takeaway: two-term recurrences need both starting values.

A growth hook: $52!$

- ▶ $52! = 80658175170943878571660636856403766975289505440883277824000000000000$
- ▶ $2^{52} = 4503599627370496$
- ▶ $52^{10} = 144555105949057024$

A shuffled deck has $52!$ possible orderings.

Start the inequality where it is true

n	0	1	2	3	4
$n!$	1	1	2	6	24
2^n	1	2	4	8	16
$n! > 2^n?$	no	no	no	no	yes

Claim: $n! > 2^n$ for every integer $n \geq 4$.

Factorial proof: Base

Base case: $n = 4$

$$4! = 24 > 16 = 2^4.$$

At $n = 0$, the proposed claim says $1 > 1$, which is false.

Factorial proof: Hypothesis

Assume for one $k \geq 4$ that $k! > 2^k$.

$$(k + 1)! = (k + 1)k!$$

Since $k + 1 > 0$, multiplying preserves the inequality.

Factorial proof: Step

$$\begin{aligned}(k+1)! &> (k+1)2^k \\ &\geq 2 \cdot 2^k \\ &= 2^{k+1}.\end{aligned}$$

Takeaway: an induction base must make the claim true.

Demo 3: odd numbers build a square

1 2 3 4
2 2 3 4
3 3 3 4
4 4 4 4

$4^2 = 1 + 3 + 5 + 7 = 16$
The next L adds 9 cells.

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2$$

Domino picture

- ▶ Base: push the first.
- ▶ Step: k knocks down $k + 1$.
- ▶ Together: every one falls.

Odd-sum proof: Base

Base case: $n = 0$

The empty sum is $0 = 0^2$.

(Or start at $n = 1$: $1 = 1^2$.)

Odd-sum proof: Hypothesis

Assume the first k odd numbers add to k^2 :

$$1 + 3 + \cdots + (2k - 1) = k^2.$$

The next odd number is $2k + 1$.

Odd-sum proof: Step

$$(1 + 3 + \cdots + (2k - 1)) + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2.$$

Takeaway: the new L has $2k + 1$ cells.

Extra: Horner's rule is a recurrence

$$p_0 = 2,$$

$$p_2 = 3p_1 + 2 = 2,$$

$$p_1 = 3p_0 - 6 = 0$$

$$p_3 = 3p_2 - 1 = 5$$

Direct check: $54 - 54 + 6 - 1 = 5$. Horner: $[2, 0, 2, 5]$.

Horner payoff

Degree	Direct multiplications	Horner multiplications
3	9	3
10	65	10

Takeaway

A recurrence can also be a faster way to compute.

Find the bug: problem

Proposed induction proof

Base: $n = 1$ works for $1 + 3 + \cdots + (2n - 1) = n^2$.

Step: Assume the first $k + 1$ odd numbers add to $(k + 1)^2$.

Remove $2k + 1$; then the first k add to k^2 . Done!

What did the proof assume, and what did it conclude?

Find the bug: reveal

The arrow points backward: it assumes $P(k + 1)$ to get $P(k)$.

The bridge we need is $P(k) \Rightarrow P(k + 1)$.

Takeaway

Check both the base case and the direction of the step.

Your turn: Tuesday activity

1. Trace $S(4)$; name the base and reducing rule.
2. For $T(0) = 1$, $T(n) = T(n+1)$, find the smallest input that exposes the problem.
3. Give the base check and bridge for $S(n) = n(n+1)/2$.

Sources $\rightarrow$ Rules $\rightarrow$ Work $\rightarrow$ Check $\rightarrow$ One-sentence summary

AI check

AI can omit a base case or reverse the induction step. Check:

1. Does the smallest input reach a true base case?
2. Does each recursive input move toward a base case?
3. Does the step assume $P(k)$ and establish $P(k + 1)$?
4. For Fibonacci, are both $F(0)$ and $F(1)$ present and correct?

Close: the bridge

- ▶ A recursive definition computes; a trace checks a case.
- ▶ Base cases start the chain.
- ▶ The inductive step carries truth from k to $k + 1$.

In one sentence: why does induction reach every case?